

Shiyue Li

Research interests: Algebraic Geometry, Combinatorics, Tropical Geometry.

EMPLOYMENT

University of Michigan , Ann Arbor, MI Donald J. Lewis Research Assistant Professor (Mentor: David Speyer)	2025–present
Institute for Advanced Study , Princeton, NJ Member (Mentor: Federico Ardila-Mantilla, June Huh)	2024–2025
Institute for Advanced Study , Princeton, NJ Member (Mentor: June Huh)	2023–2024

EDUCATION

Brown University , Providence, RI Ph.D. and M.Sc. in Mathematics (Advisor: Melody Chan) Thesis: K -rings and Hodge structures in graphs, matroids, and moduli spaces of curves	2019–2023
Yale University , New Haven, CT M.Sc. in Mathematics, Lang Fellow	2017–2019
Harvey Mudd College , Claremont, CA B.Sc. in Mathematics with Honors and Distinction (Advisor: Dagan Karp) Thesis: Chow rings of heavy/light Hassett spaces via tropical geometry	2013–2017

PREPRINTS & PUBLICATIONS

Magic positivity of Snapper polynomials for matroids
arXiv:2609.24917

Matroid analogues of Gal's conjecture
(with Basile Coron, Luis Ferroni)
arXiv:2604.04550

Structural properties of nested set complexes
(with Basile Coron, Luis Ferroni)
arXiv:2601.05188

Chow polynomials of rank-uniform labeled posets
(with Basile Coron, Luis Ferroni)
arXiv:2511.13819

Kapranov degrees

(with Joshua Brakensiek, Christopher Eur, Matt Larson)
International Mathematics Research Notices. 2025 (**20**)

Multimatroids and rational curves with cyclic action

(with Emily Clader, Chiara Damiolini, Christopher Eur, Daoji Huang.)
International Mathematics Research Notices. 2024 (**12**)

K -rings of wonderful varieties and matroids

(with Matt Larson, Sam Payne, Nicholas Proudfoot.)
Advances in Mathematics **441**. (2024)

Equivariant log-concavity of independence sequences of claw-free graphs

Séminaire Lotharingien de Combinatoire **89B**, 6. (2023)

Equivariant log-concavity of graph matchings

Algebraic Combinatorics **6**(3) (2023)

Relative Bott–Samelson varieties

La Matematica **2** (2023)

Topology of tropical moduli spaces of weighted stable curves in higher genus

(with Siddarth Kannan, Stefano Serpente, Claudia Yun)
Advances in Geometry, **23**(3) (2023)

Wonderful compactifications and rational curves with cyclic action

(with Emily Clader, Chiara Damiolini, Rohini Ramadas)
Forum of Mathematics, Sigma **11** (2022)

Permutohedral complexes and rational curves with cyclic action

(with Emily Clader, Chiara Damiolini, Daoji Huang, Rohini Ramadas)
manuscripta mathematica (2022)

Intersecting psi-classes on tropical Hassett spaces

(with Marvin Anas Hahn)
Combinatorial Theory **2**(3) (2022)

Chow ring of heavy/light Hassett spaces via tropical geometry

(with Siddarth Kannan, Dagan Karp)
Journal of Combinatorial Theory, Series A **178C** (2021)

FELLOWSHIPS, GRANTS & AWARDS

NSF Grant DMS-1926686, Institute for Advanced Study	2023–2025
Coline M. Makepeace Fellowship, Brown University	2021–2022
Robert Willets Carle Scholarship, Yale University	2018–2019
Kenneth and Mary Wang Fellowship, Yale University	2017–2018
Lang Fellowship, Department of Mathematics, Yale University	2017–2019

Harvey S. Mudd Merit Award (annually), Harvey Mudd College	2013–2017
Yu Yuen Kit So International Scholarship, Harvey Mudd College	2013–2017
Dean’s List, Harvey Mudd College (all semesters)	2013–2017
China National Linguistics Olympiad 3rd Place	2013
China National Linguistics Olympiad Team 2nd Place	2013

INVITED TALKS

LG&TBQ Bielefeld, Germany	Aug 2027
JMM-AWM Special Session on “ <i>WIAG: Topics in Algebraic Geometry</i> ” Chicago	Jan 2027
Workshop on “ <i>Enumerative Geometry of Matroids</i> ” Oberwolfach, Germany	Dec 2026
ALGECOM University of Illinois Urbana-Champaign	Nov 2026
<i>Combinatorics Seminar</i> University of Michigan	Sep 2026
Workshop on “ <i>Algebraic Combinatorics in Pisa</i> ” Università di Pisa	Jun 2026
Workshop on “ <i>Algebra and Geometry of Matroids</i> ” Banff International Research Station of Casa Matemática Oaxaca, Mexico	Sep 2026
Institut des Sciences mini-course on “ <i>Geometry and Combinatorics of Moduli Spaces</i> ” Université du Québec à Montréal	May 2026
<i>Algebraic Geometry Seminar</i> University of Michigan	Apr 2026
AMS Special Session on “ <i>Geometry of Moduli Spaces</i> ” Boston College	Mar 2026
<i>Algebraic Geometry Seminar</i> University of Pennsylvania	Apr 2025
AMS Special Session on “ <i>Cohomology of Arithmetic Groups, Mapping Class Groups, and Moduli Spaces</i> ” Jan 2025 Joint Mathematics Meetings, Seattle, USA	
Workshop on “ <i>Geometry of Matroids</i> ” Institute for Advanced Study	Oct 2024
“ <i>What is...?</i> ” Seminar Institute for Advanced Study	Oct 2024
Workshop on “ <i>Moduli spaces of curves, moduli spaces of graphs and graph complexes</i> ” ETH Zürich, Switzerland	Sep 2024
Workshop on “ <i>Matroid Theory</i> ” Suzhou University, Suzhou, China	Jul 2024

Workshop on “ <i>Arrangements, Matroids and Logarithmic Vector Fields</i> ”, Oberwolfach, Germany	Jun 2024
<i>Algebraic Geometry Seminar</i> Columbia University	Apr 2024
<i>Algebra Seminar</i> Tufts University	Apr 2024
“ <i>What is...?</i> ” <i>Seminar</i> Institute for Advanced Study	Mar 2024
<i>Mathematical Conversations</i> Institute for Advanced Study	Mar 2024
<i>Combinatorics Seminar</i> (online) University of Minnesota	Feb 2024
<i>F₁ World Seminar</i> (online)	Jan 2024
Special Session on “ <i>Combinatorial perspectives on algebraic curves and their moduli</i> ” Joint Mathematics Meetings, San Francisco, USA	Jan 2024
Preparatory Lectures for <i>Chow Lectures</i> Max Planck Institute, Leipzig, Germany	Oct 2023
Workshop on “ <i>Curves: Algebraic, Tropical, and Logarithmic</i> ” Banff International Research Station, Canada	Aug 2023
<i>The 35th Formal Power Series and Algebraic Combinatorics (FPSAC 2023)</i> University of California, Davis	Jul 2023
Workshop on “ <i>GEMS in Combinatorics</i> ” American Institute of Mathematics, USA	Mar 2023
Workshop on “ <i>Algebraic Aspects of Matroids</i> ” Banff, Canada Banff International Research Station, Canada	Mar 2023
Lecture series on “ <i>Growing up in Sciences</i> ” Bowdoin College	Feb 2023
Association of Women in Mathematics Student Seminar (online) University of Pittsburgh	Feb 2023
<i>Combinatorics Seminar</i> University of Michigan	Feb 2023
<i>Matroid Seminar</i> (online) Fields Institute	Dec 2022
<i>Algebra and Algebraic Geometry Seminar</i> University of Washington	Nov 2022
<i>MIT-Harvard-MSR Combinatorics Seminar</i> Harvard University	Nov 2022
AMS Special Session on “ <i>Combinatorial Algebraic Geometry</i> ” University of Massachusetts, Amherst	Oct 2022
<i>AWM Research Symposium</i> University of Minnesota	Jun 2022
<i>RATCOW Seminar</i> University of Oregon	Jun 2022

<i>BATMOBILE</i> Brown University	Jun 2022
<i>Combinatorial and Nonlinear Day</i> Brown University	Apr 2022
<i>Joint Mathematics Meetings AWM Special Session (online)</i>	Apr 2022
<i>Canadian Math Society: Combinatorics and Geometry of Moduli Spaces (online)</i>	Dec 2021
<i>Front Range Algebraic Geometry and Number Theory</i> Colorado State University	Sep 2021
<i>SIAM Conference on Applied Algebraic Geometry (online)</i>	Aug 2021
<i>Moduli Across the Pandemic (online)</i>	Apr 2021
<i>Tropical Geometry in Frankfurt (TGiF) Seminar (online)</i>	Apr 2021
AMS Spring Eastern Sectional Meeting on “Moduli of Curves” (online)	Mar 2021
<i>Algebra Seminar (online)</i> Brown University	Nov 2020
<i>American Graduate Student Algebraic Geometry Seminar (online)</i>	Oct 2020

SERVICE

Referee for

Advances in Mathematics
 Algebraic Combinatorics
 Duke Mathematical Journal
 European Journal of Combinatorics
 Forum of Mathematics, Sigma ($\times 2$)
 International Mathematics Research Notices ($\times 3$)
 Journal of the London Mathematical Society ($\times 3$)
 Journal für die reine und angewandte Mathematik (Crelle’s Journal)*
 Mathematische Zeitschrift[†]
 Proceedings of the American Mathematical Society ($\times 2$)
 Proceedings of the London Mathematical Society
 Séminaire Lotharingien de Combinatoire (Proceedings of FPSAC Conferences)[†] ($\times 3$)

(* means quick opinions only; † means quick opinion and/or detailed report)

Organization

University of Michigan AWM Advisory Board	2025–2026
University of Michigan Combinatorics Lunch, co-organizer	2025–
IAS Graph Complexes Learning Seminar, Organizer	Fall 2024
IAS School of Mathematics Monthly Women’s Lunch, co-organizer	2023–2025
IAS “What is...?” Seminar, Co-organizer	2023–2025
AMS Student Chapter at Brown University, Chair	2019–2021
Diversity and Inclusiveness Committee at Brown University, Member	2019–2020
Directed Reading Program at Brown University, Organizer	2019–2021
Directed Reading Program at Yale University, Co-organizer with Fernando Al Assal	2017–2018
Intersection: Womxn in Mathematics at Yale University, Co-organizer	2018
Yale Graduate and Professional Student Senate, Senator	2017–2018
Harvey Mudd College Math Club, President	2016–2017

TEACHING

University of Michigan, Instructor

Math 565 Combinatorics and Graph Theory (Graduate)	Fall 2026
Math 566 Combinatorial Theory (Graduate)	Winter 2026
Math 275 Introduction to Cryptography (Inquiry-Based Learning)	Fall 2025

Brown University Sheridan Teaching Certificate I

Fall 2021

Brown University, Teaching Assistant

Math 90 Introduction to Calculus	Spring 2023
Math 100 Calculus Part II	Fall 2022
Math 100 Calculus Part II	Spring 2022
Math 100 Calculus Part II	Spring 2021
Math 100 Calculus Part II	Fall 2020

Canada/USA Mathcamp, Mentor

Summer 2019

Introduction to Group Theory
Young tableaux and Representation Theory
Young tableaux and Combinatorics
Young tableaux and Probability

Canada/USA Mathcamp, Mentor

Summer 2018

Algebraic Number Theory
Geometric Group Theory
Introduction to Modular Forms
Tropical Plane Curves
Rational Points on Elliptic Curves

Yale University, Teaching Assistant

Math 225 Linear Algebra and Matrix Theory	Fall 2018
Math 112 Single Variable Calculus	Spring 2018

MENTORING

LoG(M) at the University of Michigan

Fall 2026

Edward (Weizhen) Tong, Co-mentor: Bailee Zacovic.
Chow functions of unbounded posets.

Reading course at the University of Michigan

Summer 2026

Yujie Guo, Kate Krotkov. Co-mentor: Sam Payne
Introduction to toric varieties

Summer Undergraduate Math Research at Yale

2020

Dawit Mengesha, Robert Miranda, Brian Sun. Co-mentor: Max Kutler.
Topological zeta functions of matroids.

Canada/USA Mathcamp Learning Projects

Summer 2019

Iliaria Seidel.
Alice Jenkins, Jeremy Zhou. *Schubert calculus.*
Reed Jacobs. *Primes of the form $x^2 + ny^2$.*
Simran Khunger, Rupert Li. *Enumerative geometry and Young tableaux*
Serena An. *Group actions and Sylow's theorems.*

Brown Mathematics Directed Reading Program	2019
Deven Carmichael, Joseph Hlavinka. <i>Ideals, Varieties and Algorithms</i> .	
Dominick Joo, Elliott Lehrer, Joshua Lebo. <i>Foundations of Algebraic Geometry</i> .	
Women in Sciences at Yale	2018
Alara Değirmenci.	

OUTREACH

Princeton Noetherian Rings Seminar	Apr 2024
Brown Math Slam	Apr 2023
Growing up in Sciences Lecture Series, Bowdoin College	Feb 2023
Summer Undergraduate Math Research at Yale (SUMRY) Colloquium	Jul 2020
Yale Undergraduate Mathematics Society (YUMS) Colloquium	Nov 2019
Department Undergraduate Group (DUG) Colloquium at Brown	Oct 2019

OTHER SCIENTIFIC POSITIONS

Google Research. Mountain View, CA.	Summer 2016
Research Intern. Mentors: Caroline Pantofaru, Malcolm Slaney	
Howard Hughes Medical Institute. Pomona, CA.	Summer 2014
Research Assistant. Mentor: Blerta Shtylla.	
Summer Science Program. Socorro, NM.	Summer 2012
Asteroid 1998QS52 Orbit Determination (Harvard–Smithsonian Center archived).	